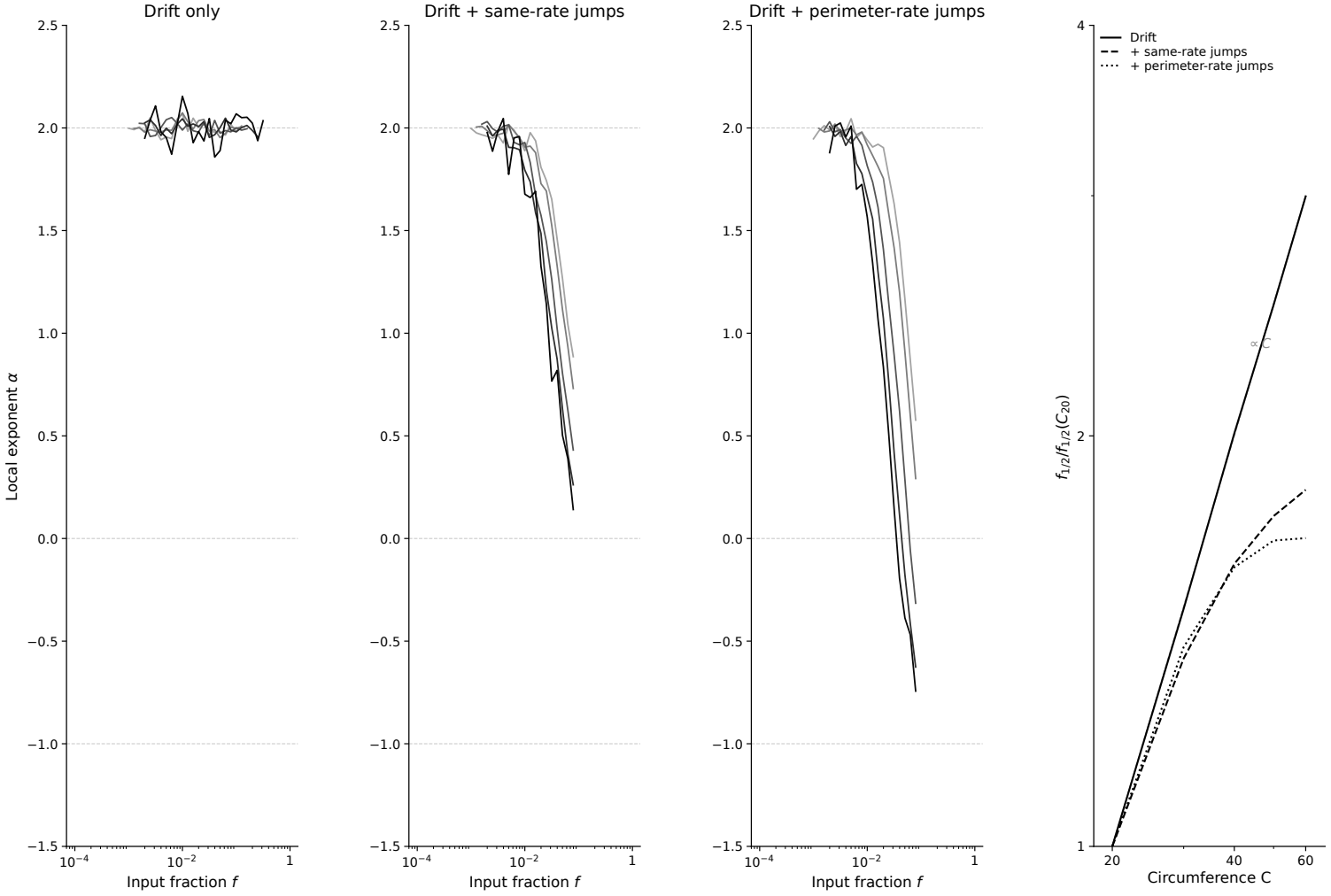
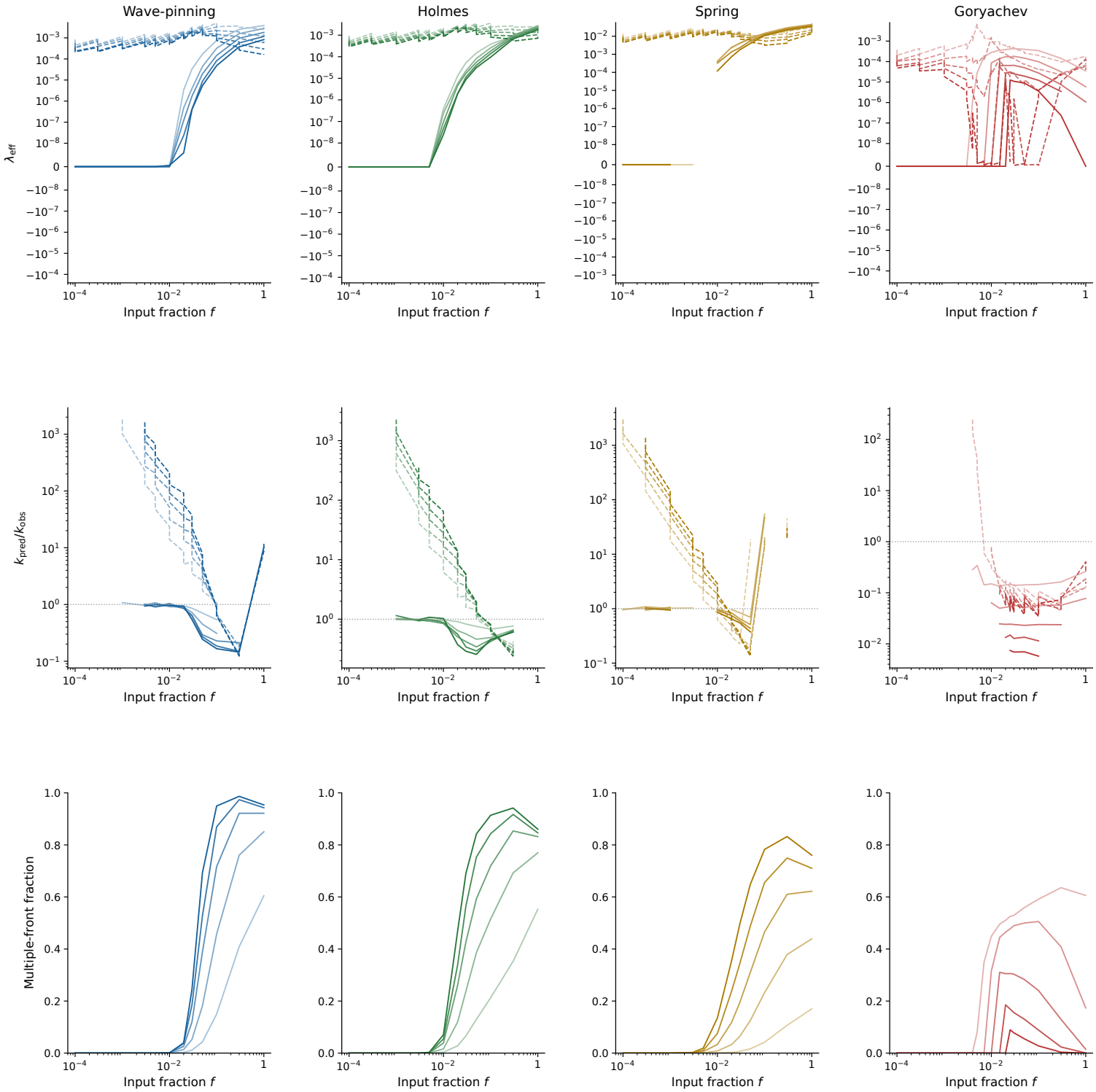
S4.1 Displaced null-mixture exponents and tolerance folds



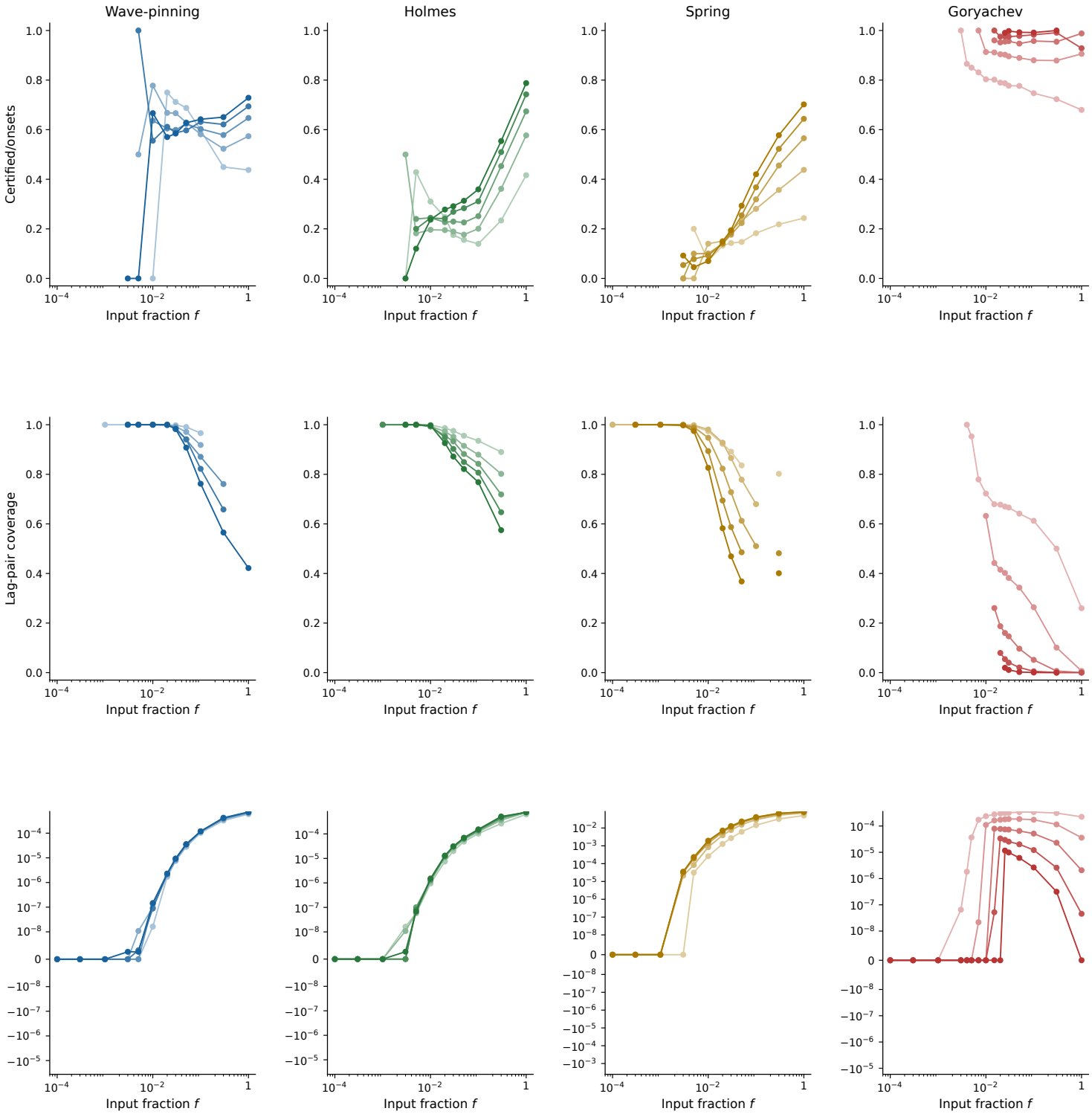
Historical simulated nulls; unchanged source records. These diagnostic curves were removed from the main figure.

S4.2 Displaced drift/relocation diagnostics



Solid: tracked; dashed: detector-free fit. All inputs and failures retained, with no rare-multiple filter. Native clocks differ across models.

S4.3 Identification and endpoint coverage



Certified births are lower bounds. All persistent secondary onsets give upper bounds including splits. Different certification fractions prevent population-level comparisons of takeover probability.

S4.4 Missing information and exact scope of the birth observable

Requested split counts are unavailable for every one of the 260 conditions.

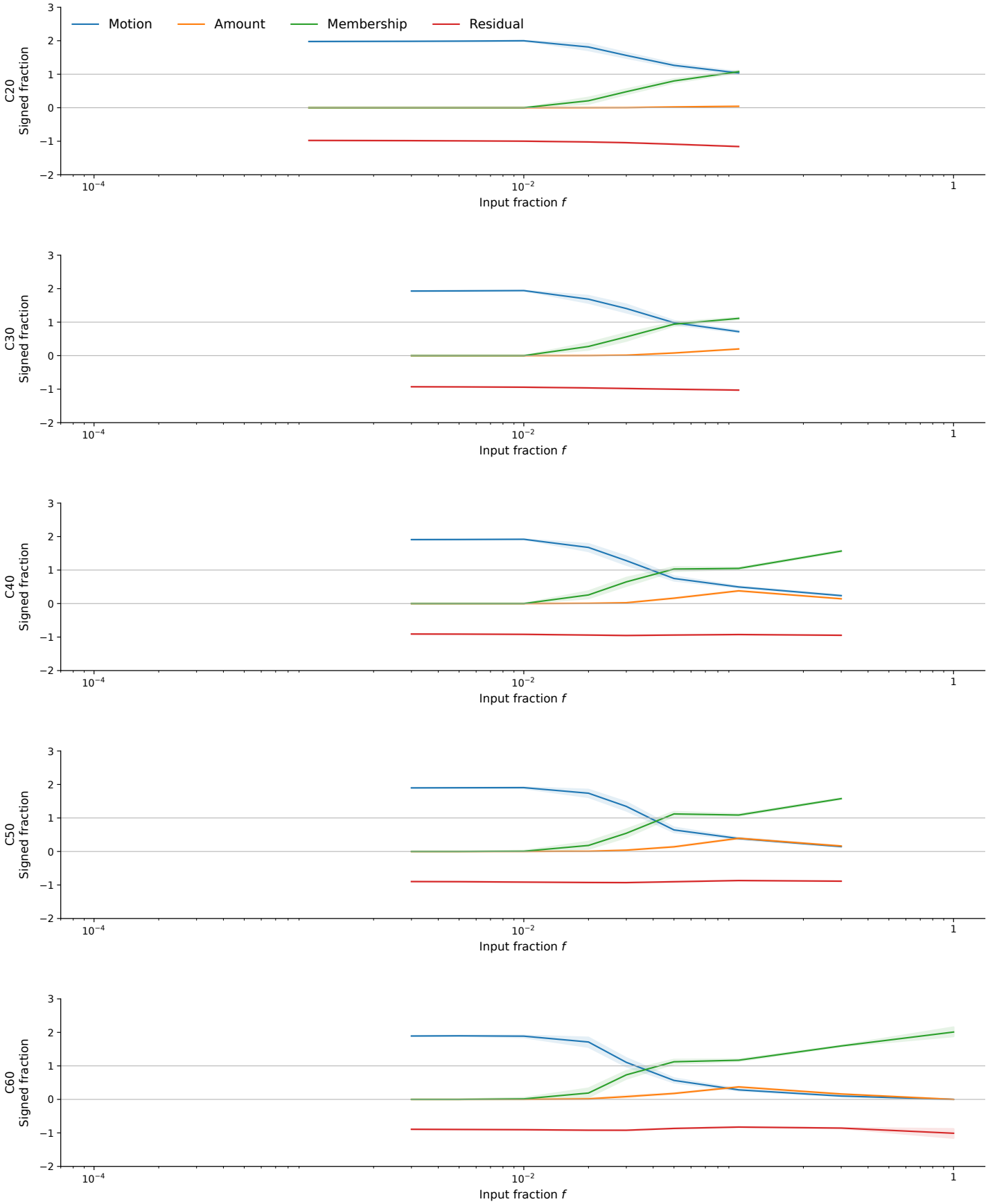
The compact producer stored maximum-overlap identities, excess-weighted contour centroids, full-basin amounts and contour-cell widths. It discarded contour masks/endpoints, parent IDs, per-region plateau heights and event lists. A centroid and width do not determine the component interval, because the centroid is weighted by the unknown within-component profile.

A certified no-parent birth is a new identity that persists for at least five frames and whose conservative angular support enclosure misses every previous enclosure. This is a subset of all no-overlap births. No uncertified track is called a split or a true birth.

Incumbent half-width is estimated as full-basin amount / (2 × current global contrast), using the prior frame. It is an explicit plateau approximation, not a recovered local plateau. Takeovers require five consecutive raw dominant-ID frames, within a common follow-up of 20 noise blocks (1,000 chemical units; 100 Spring). Shorter follow-up is excluded. Distances are grouped as $d \le 0$, $0 < d \le C/6$ and $d > C/6$; designed inputs $f > 0.01$ are pooled.

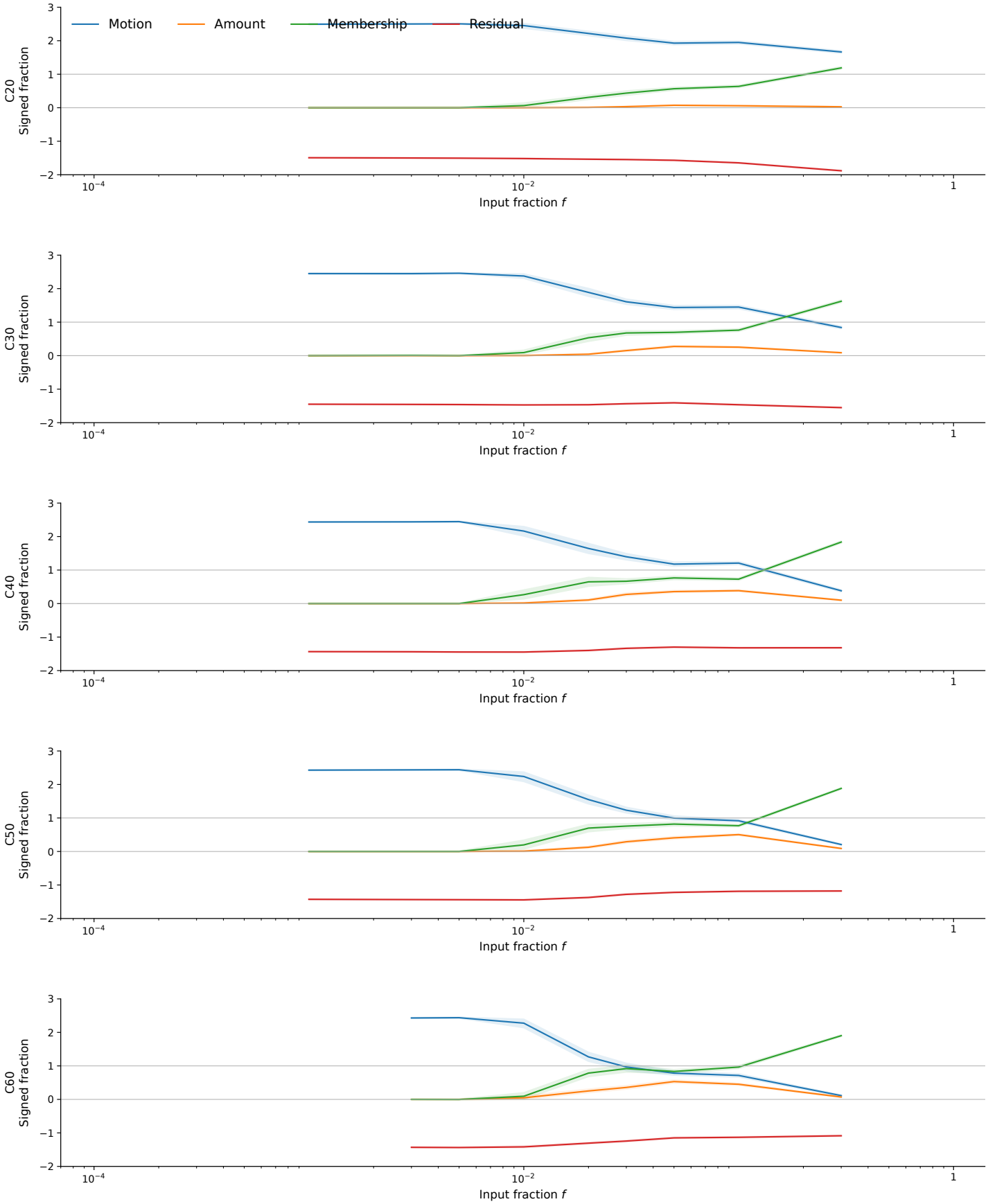
The historical aggregate region audits used a two-frame pairwise-dominance rule and include split onsets. They cannot replace the new five-frame event definition. No noise runs were added.

S4.5 Wave-pinning: signed heading-loss fractions



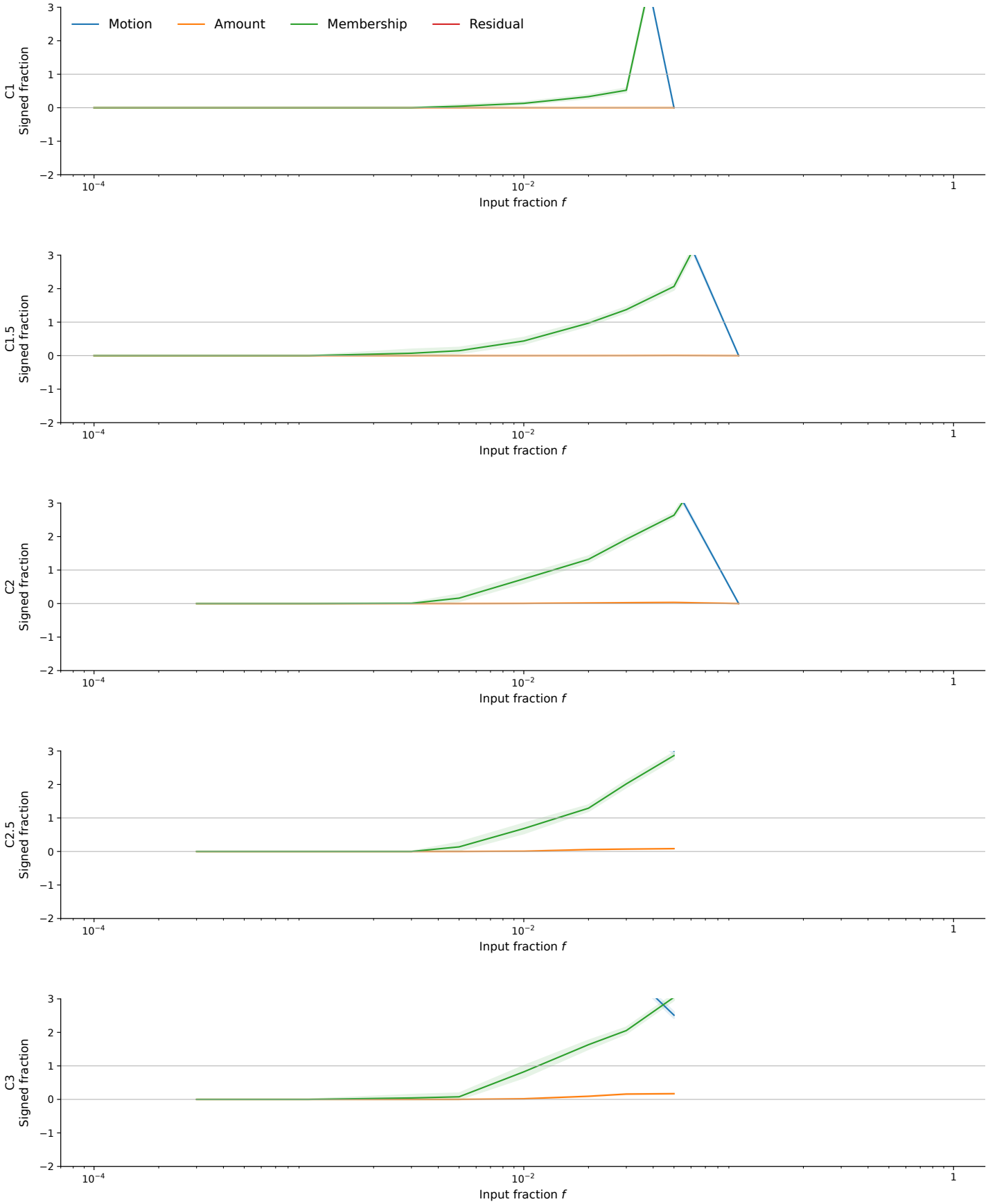
Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

S4.5 Holmes: signed heading-loss fractions



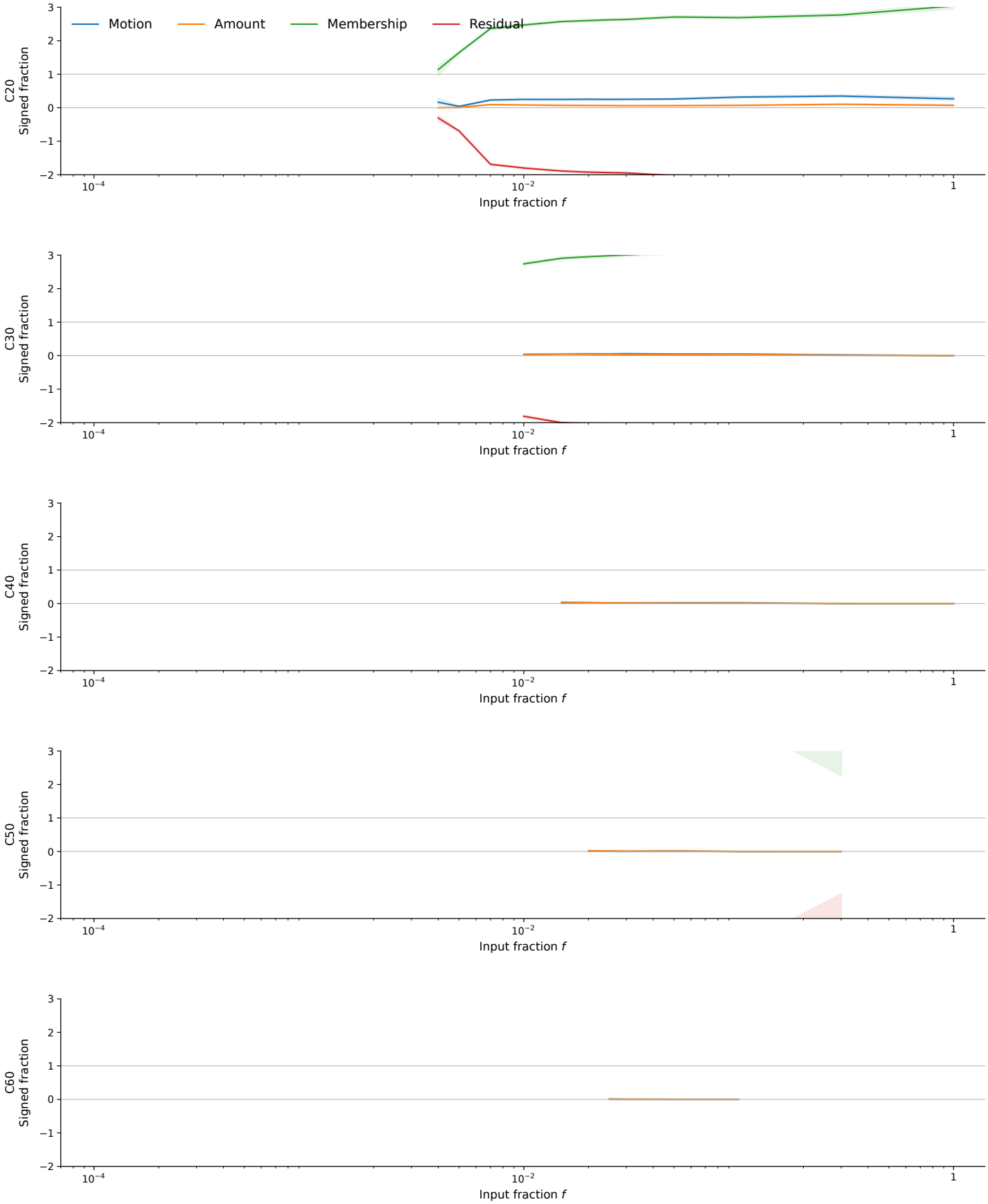
Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

S4.5 Spring: signed heading-loss fractions



Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

S4.5 Goryachev: signed heading-loss fractions



Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.